

AUTOMATIC NUMBER PLATE RECOGNITION SYSTEM

A Project Report
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in

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

Under the guidance of
MS. PARVATHY JYOTHI



CREATING TECHNOLOGY
LEADERS OF TOMORROW
ESTD 2002

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA
SCIENCE



Jyothi Engineering College
Reaccredited with NAAC (Grade A) and NBA Programmes*

Approved by AICTE and Affiliated to APJ Abdul Kalam Technological University
A CENTRE OF EXCELLENCE IN SCIENCE AND TECHNOLOGY BY THE CATHOLIC ARCHDIOCESE OF TRICHUR
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*NBA accredited B.Tech Programmes in Civil Engineering, Computer Science and Engineering, Electronics and Communication Engineering, Electrical and Electronics Engineering and Mechanical Engineering valid till 2025, Mechatronics Engineering valid till 2026

April 2025

DECLARATION

We hereby declare that the project report “ AUTOMATIC NUMBER PLATE RECOGNITION SYSTEM”, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of MS. PARVATHY JYOTHI. This submission represents the ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in this submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously used by anybody as a basis for the award of any degree, diploma or similar title of any other University.

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CERTIFICATE

This is to certify that the report entitled “ AUTOMATIC NUMBER PLATE RECOGNITION SYSTEM ” submitted by GOUTHAM KRISHNA(JEC21AD023) GOWRIPRIYA(JEC21AD024)NANDANA ANAND(JEC21AD036) SHANE SUNNY(JEC21AD044) to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree in Bachelor of Technology in Artificial Intelligence And Data Science is a bonafide record of the project work carried out by them under my/our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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- PEO 1: Graduates will demonstrate comprehensive expertise in artificial intelligence, machine learning, and data analysis while applying interdisciplinary knowledge in computer science, statistics, and advanced mathematics to address real-world challenges.
- PEO 2: Graduates will innovate and create impactful solutions ethically, contributing to societal betterment, leveraging their leadership, collaboration, and communication skills for effective teamwork and multidisciplinary endeavors.
- PEO 3: Continuously fostering a culture of lifelong learning, graduates will adapt to evolving technological landscapes, demonstrating adaptability and a commitment to ongoing professional development in the dynamic field of AI and data science.

PROGRAM SPECIFIC OUTCOMES

The students upon completion of Program, will be able: -

- PSO1: Graduates will be able to apply core principles of science, engineering, and mathematics to develop AI-driven solutions for real-world challenges.
- PSO 2: Graduates will design, develop and implement efficient as well as scalable algorithms using programming languages, data structures and cloud computing technologies.
- PSO 3: Graduates will integrate ethical, social, and security considerations while developing AI-driven intelligent systems, ensuring responsible AI deployment.

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1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

COURSE OUTCOMES

COs	Description
CO.1	Identify technically and economically feasible problems of social relevance.
CO.2	Identify and survey the relevant literature for getting exposed to related solutions.
CO.3	Perform requirement analysis and identify design methodologies and develop adaptable and reusable solutions of minimal complexity by using modern tools and advanced programming techniques.
CO.4	Prepare technical report and deliver presentation.
CO.5	Apply engineering and management principles to achieve the goal of the project.

CO MAPPING TO POs

COs	POs											
	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	P012
CO.1	3	3	3	3	0	3	3	2	3	3	3	3
CO.2	2	1	3	2	2	2	0	2	2	2	2	2
CO.3	3	1	2	1	1	1	2	3	3	3	3	3
CO.4	2	2	2	2	2	0	0	1	2	1	2	2
CO.5	3	2	3	2	2	3	3	2	3	0	2	2
Average	2.6	1.8	2.6	2	1.4	1.8	1.6	2	2.6	1.8	2.4	2.4

CO MAPPING TO PSOs

COs	PSOs		
	PS01	PS02	PS03
CO.1	3	3	3
CO.2	3	3	2
CO.3	2	2	1
CO.4	3	2	3
CO.5	3	2	3
Average	2.8	2.4	2.4

ABSTRACT

Traffic sign recognition plays a crucial role in ensuring road safety by providing real-time information to drivers. The project uses CNN machine learning algorithm to accurately detect and classify traffic signs from images. It could help to enhance driver awareness by identifying various traffic signs. The project has some major steps, the first step is pre-processing, where the input images are transformed to improve the quality for training the machine learning model. This includes operations like resizing and cropping to ensure consistency and reduce noise. The implementation of a Convolutional Neural Network (CNN) architecture involves utilizing 2D max-pooling and 2D convolutional layers, with the activation functions being ReLU and Softmax, respectively. The model is trained on a training dataset of labelled traffic sign images. Pre-processed images along with their corresponding labels are fed into the network, enabling the model to extract intricate features and information from the images. During training, the network's parameters are optimized using back propagation algorithm to minimize the error between predicted and actual classes. Once the training is complete, the model is evaluated using a test dataset. This measures the accuracy, precision and recall of the model to recognize different types of signs. Finally, the trained model can be deployed in real-world applications. It can take input from images, perform recognition, and provide information to drivers in real-time. The system's performance can be updated to ensure its effectiveness. This project provides a practical solution for automating the detection and classification of traffic signs. It enhances road safety, improves driver awareness and contributes to the development of autonomous vehicles.

CONTENTS

List of Figures	xiii
1 Introduction	1
1. Overview..... Objectives	1
1 Organization of the	1
1. Project	2
2 Literature Survey	3
2. AutomaticNepalnumberplaterecognitionwithsvm	3
3 AutomaticCarNumberPlaterecognition Vehicle Number	3
2. Plate Recognition System: Implementation using Template	
2 Matching..... Cognitive number plate recognition using	4
2.4 machine learning and data visualization	
3 techniques	4
2.5 Automatic License Plate Recognition Using Image Processing And Neural	
Network	4
2.6 Automatic Number Plate Recognition:A Detailed Survey of Relevant Algorithms	
.	5
2. Automatic number platerecognition using deep	5
7 A machine learning algorithm to automate v	
2. platedetection	6
3 Methodology	7
3.1 ExistingSystems	7
3.1.1 Disadvantagesofexistingsystems	7
3. Problem Statement	7
2 ProposedSystem.....	7
3. Requirement Analysis	8
3 3.4.1Functional.....	8
3. 3.4.2 NonFunctional	9
4.5 Modules.....	10

3.5.1 Datacollectionandpreprocessing ..	1
3.5.2 DataSplitting.....	0
3.5.3 ModelArchitecture.....	1
3.5.4 Modeltraining	1
3.5.5 ModelEvaluation.....	1
3. System requirements and specifications	1
6 DataFlowDiagram	1
3. 3.7.1 ImplementationFlowChartDiagram.	1
3.8 Implementation.....	1
3.8.1 ArchitectureDiagram	1
3.8.2 ImportingLibraries	1
4 Results&Discussion	1
4. Result ...	3
1 Discussion	1
5 Conclusion&FutureScope	3
8. Conclusion .	1
1 Future Scope	3
5References	1
2Appendices	5
	1
	5
	17
	1
	7
	2
	1
	2
	2
	2
	2
	2
	2
	2
	4
	2

LIST OF FIGURES

Figure No.	Title	Page No.
3.1	SampleImagesfromthedataset(indiancarnumberplate).	1
3.2	FlowChart.....	0
3.3	Architecturediagram	1
4.1	TrainLoss.....	3
4.2	ValidationLoss	1
4.3	Confusionmatrix	5
4.4	OpeningInterface	1
4.5	testing.....	8
		1
		8
		1
		9
		2
		0
		2
		0

CHAPTER 1

INTRODUCTION

1.1 Overview

Indoor navigation within academic institutions has become an increasingly important challenge due to the growing size and complexity of campus buildings. New students, visitors, and staff often face difficulties in locating classrooms, laboratories, offices, and other facilities, especially in unfamiliar environments. Unlike outdoor navigation, indoor spaces cannot rely on GPS due to weak or unavailable signals. As a result, traditional aids such as printed maps and static signboards remain in use, despite their limited ability to provide real-time, personalized guidance. Existing indoor navigation systems often depend on hardware-based technologies such as Wi-Fi fingerprinting, Bluetooth beacons, or sensor networks, which increase deployment cost, maintenance effort, and system complexity, making them unsuitable for many academic environments.

To address these limitations, this project proposes a QR-based indoor navigation system integrated with an AI-powered chatbot. The system represents the indoor environment as a two-dimensional grid derived from the building's floor plan, where walkable and non-walkable areas are clearly defined. An efficient pathfinding algorithm computes the shortest route between locations. Users interact with the system through a conversational chatbot by specifying their current location and destination using natural language. QR codes placed at strategic locations provide instant access to the web-based interface without requiring additional applications or specialized hardware. This lightweight, cost-effective approach enhances usability and offers an intuitive solution for indoor campus navigation.

1.2 Objectives

The primary objective of this project is to design and develop an intelligent indoor navigation system that assists users in efficiently navigating academic buildings without relying on GPS or hardware-intensive positioning technologies. The system aims to provide accurate, reliable, and user-friendly navigation support for students, visitors, and staff within campus environments by leveraging software-based solutions.

Another key objective is to implement a QR-based access mechanism that allows users to enter the navigation system instantly through a simple scan, eliminating the need for dedicated mobile applications or additional infrastructure. This ensures ease of deployment and accessibility across different devices, including smartphones and tablets, while reducing maintenance complexity and cost.

The project also aims to integrate an AI-powered conversational chatbot to enhance user interaction. Instead of traditional menu-driven navigation, the chatbot enables users to specify their current location and destination using natural language. This objective focuses on improving usability, reducing user effort, and making the navigation experience more intuitive and interactive, particularly for first-time users.

A further objective is to model the indoor environment using a grid-based representation of the floor plan, clearly distinguishing walkable and non-walkable areas. This structured spatial modeling enables efficient path computation using a heuristic-based pathfinding algorithm, ensuring that the shortest and most optimal routes are generated accurately.

Additionally, the project aims to visualize the computed navigation path clearly on a digital floor plan, highlighting the source, destination, and route for easy interpretation. Finally, the system is designed with scalability in mind, allowing future enhancements such as multi-floor navigation, improved natural language understanding, real-time location tracking, and augmented reality-based guidance.

1.3 Organization of the Project

The report is organised as follows:

- Chapter 1: Introduction- outlines smart indoor navigation and a chatbot-based approach
- Chapter 2: Literature Survey- Summarizes the various existing techniques that helped us in achieving the desired result.
- Chapter 3: Methodology- Methods which are used in this project.
- Chapter 4: Results and Discussion- The results of work and discussion
- Chapter 5: Conclusion & Future Scope- The chapter gives a conclusion of the overall work along with the future scope of implementation.
- Chapter 6: References- Includes the references for the project.

CHAPTER 2

LITERATURESURVEY

2.1 Hybrid Campus Navigation System: Seamless Integration of Indoor and Outdoor Routing,

Ramamoorthy, Yadav, and Pal proposed a Hybrid Campus Navigation System that integrates both indoor and outdoor routing to provide seamless navigation across a university campus. Their system addresses the limitation of conventional navigation solutions that handle indoor and outdoor spaces separately. Outdoor navigation is supported using GPS-based routing, while indoor navigation relies on digital floor maps and predefined pathways. The integration of these two navigation modes allows users to transition smoothly between outdoor campus areas and indoor buildings without interruption. The authors emphasize accuracy, continuity, and user convenience as the core strengths of their approach. The system demonstrates improved navigation efficiency across large campus environments by combining multiple navigation technologies. However, the proposed solution requires careful coordination between indoor maps and outdoor positioning systems. This work highlights the importance of unified navigation systems and provides valuable insights that motivate the development of lightweight, scalable campus navigation solutions

2.2 Survey of wireless indoor positioning techniques and systems

Liu, Darabi, Banerjee, and Liu presented a comprehensive survey of wireless indoor positioning techniques and systems, highlighting the challenges of accurate localization in indoor environments. The study reviews various technologies including Wi-Fi fingerprinting, Bluetooth, RFID, infrared, and ultra-wideband systems. The authors analyze the strengths and limitations of each method in terms of accuracy, cost, scalability, and infrastructure requirements. While many techniques achieve high positioning accuracy, they often depend on extensive hardware deployment, signal calibration, and maintenance. The survey emphasizes that environmental changes such as walls, obstacles, and signal interference significantly affect system performance. This work provides a foundational understanding of wireless indoor positioning systems and underscores the trade-offs between accuracy and deployment complexity. The findings motivate the need for alternative, software-centric indoor navigation approaches that reduce hardware dependency while maintaining usability and reliability.

2.3 A formal basis for the heuristic determination of minimum cost paths

Hart, Nilsson, and Raphael introduced the A* algorithm, providing a formal and systematic approach for determining minimum-cost paths using heuristic guidance. Their work combines actual path cost with an estimated cost to the goal, enabling efficient search while ensuring optimality under Hart, Nilsson, and Raphael introduced a foundational work that provides a formal basis for the heuristic determination of minimum-cost paths, leading to the development of the well-known A* search algorithm. This research establishes a systematic approach for pathfinding by combining the actual cost from the start node with a heuristic estimate of the remaining cost to reach the goal. The authors demonstrate how admissible and consistent heuristics ensure optimality and efficiency in path computation. By guiding the search toward the most promising paths, the algorithm significantly reduces computational complexity compared to uninformed search methods.

The paper highlights the applicability of heuristic-based path planning in structured environments, where movement costs and constraints can be clearly defined. The A* algorithm has since become a cornerstone in artificial intelligence, robotics, and navigation systems due to its balance between accuracy and performance. In the context of indoor navigation, this work is particularly relevant as grid-based representations align well with the algorithm's assumptions. The principles introduced in this study enable efficient route planning in environments with obstacles and restricted areas. This research forms the theoretical foundation for modern indoor navigation systems that rely on heuristic pathfinding for optimal route generation. heuristics. The study established the theoretical foundation for heuristic search methods and demonstrated their effectiveness in graph-based navigation problems. A* has since become a widely adopted algorithm in navigation, robotics, and path planning applications due to its accuracy and computational efficiency.

2.4 A note on two problems in connexion with graphs

Dijkstra introduced a fundamental algorithm for solving shortest path problems in weighted graphs, which later became known as Dijkstra's algorithm. This work focuses on determining the minimum-cost path between nodes in a graph where all edge weights are non-negative. The algorithm systematically explores nodes by selecting the one with the smallest known distance from the source, ensuring that the shortest path is found efficiently. Due to its deterministic nature and guaranteed optimality, Dijkstra's algorithm has been widely used in network routing, transportation systems, and navigation applications.

The study provides a strong theoretical foundation for graph-based path planning and demonstrates how structured representations of space can be leveraged for optimal route computation. However, the algorithm explores paths uniformly without using heuristics, which can increase computational cost in large or dense environments. Despite this limitation, Dijkstra's algorithm remains highly reliable and accurate. In indoor navigation systems, it serves as an important baseline for comparison with heuristic-based approaches such as A*. This work significantly influenced the development of modern pathfinding techniques and continues to be relevant in both theoretical and practical navigation systems.

2.5 Artificial Intelligence: A Modern Approach

Russell and Norvig's *Artificial Intelligence: A Modern Approach* is a comprehensive and widely accepted reference that presents fundamental concepts, techniques, and applications of artificial intelligence. The book provides detailed explanations of intelligent agents, search algorithms, heuristic methods, and problem-solving strategies that form the core of many AI-based systems. It extensively discusses classical search techniques such as uninformed and informed search, including algorithms like A* and heuristic evaluation functions. The authors emphasize the importance of rational decision-making and efficient problem-solving in complex environments. This work also explores knowledge representation, reasoning, and learning techniques that support intelligent system behavior. In the context of indoor navigation, the concepts presented in this book provide a strong theoretical foundation for heuristic-based pathfinding and intelligent interaction mechanisms. The principles discussed by Russell and Norvig support the integration of AI-driven reasoning with practical navigation systems, making this reference highly relevant to intelligent indoor navigation and chatbot-based applications.

2.6 CrowdInside: Automatic construction of indoor floorplans

Alzantot and Youssef proposed CrowdInside, a novel system for the automatic construction of indoor floor plans using data collected from crowd-sourced smartphone sensors. The system leverages inertial sensors, such as accelerometers and gyroscopes, to track user movement and infer indoor structural features including corridors, rooms, and intersections. By aggregating data from multiple users, CrowdInside is able to generate accurate indoor maps without requiring pre-existing floor plans or manual surveying. This approach reduces the cost and effort associated with creating indoor layouts, making it scalable for large buildings. However, the system relies heavily on continuous sensor data collection and user participation, which may introduce noise and accuracy issues. In the context of indoor navigation, this work highlights an alternative method for obtaining indoor spatial representations. While effective for map generation, it differs from grid-based, static-map approaches that prioritize simplicity and low infrastructure requirements.

2.7 Chatbot for university related FAQs

Ranoliya, Raghuwanshi, and Singh presented a chatbot system designed to handle university-related frequently asked questions, aiming to improve access to administrative and academic information. The proposed chatbot uses natural language processing techniques to understand user queries and provide relevant responses related to admissions, courses, examinations, schedules, and general campus information. By automating responses to repetitive queries, the system reduces the workload on administrative staff and improves response time for students. The study demonstrates how conversational interfaces can enhance user interaction and information retrieval in educational institutions. However, the chatbot is primarily limited to textual information delivery and does not support spatial navigation or location-based assistance. Despite this limitation, the work highlights the effectiveness of chatbots in academic environments and establishes the importance of natural language interaction for user engagement. This research motivates the integration of chatbot technology with indoor navigation systems, where conversational interfaces can be extended beyond information retrieval to guide users spatially within campus buildings.

CHAPTER 3

METHODOLOGY

3.1 Existing Systems

Existing system does not use the capabilities of artificial intelligence. Here licence plate recognition is done by manual interpretation. This method is prone to error because it can cause misjudgements. There are chances of human error in this process, especially during harsh weather conditions. The wrong interpretations of numbers from number plates can mislead the officials in case of law enforcement and traffic management. As there is notable surge in the number of vehicles in the road, the role of an effective number plate recognition system is very crucial.

3.1.1 Disadvantages of existing systems

1. Sometimes obstacles and inclement weather can reduce the effectiveness of automatic license plate recognition systems. When this occurs, it may be necessary to turn off security measures and increase the need for manned surveillance.
2. There are privacy concerns regarding the keeping and storing of records and images. Most commonly, people worry that the location data contained in all of this video could be exploited. It may fall prey to data thieves or individuals with other nefarious goals.
3. Automated systems that recognize license plates still have a number of issues. The most frequent issue they deal with is the variation in license plate number models across various cities and nations. They might also differ in length. The software needs to be adjusted to the location where it is being used for this reason.

3.2 Problem Statement

To develop an automatic number plate recognition system capable of detecting license plates from image/video streams, extracting their numbers, mitigating challenges like obscured plates and varying lighting conditions, and minimizing manual interventions to reduce the potential for human errors.

3.3 Proposed System

The proposed system is an Automatic Number Plate Recognition System that utilizes deep learning techniques, specifically You Only Look Once(YOLO) and Optical Character Recognition(OCR)[9], to automatically detect and extract characters from number plates. The system takes in input as images/videos of vehicles, processes them through a pre-trained

YOLOv8 model, and outputs the detected number plates. This output is then passed to an EasyOCR model to effectively extract the characters from the detected number plates. The system aims to provide accurate number plate detection and recognition, aiding in various applications such as traffic management, Law enforcement, and parking. By utilising advanced computer vision algorithms and machine learning, the proposed system offers a reliable and efficient solution for licence plate recognition, contributing to enhanced road safety and efficient traffic management.

3.4 Requirement Analysis

3.4.1 Functional

- **Training the Model:** The program should allow the user to train a YOLO model using a given image dataset.
- **Saving the Model:** The program should provide an option for the user to save the trained model to a file for future use.
- **Evaluating the Model:** The program should enable the user to evaluate the performance of the trained model using a separate test dataset. It should provide accuracy and confusion matrix as output.
- **Visualizing Training:** The program should allow the user to visualize the accuracy and loss curves during the model training process.
- **Loading a Pre-trained Model:** The program should allow the user to load a pre-trained model from a file for further use.
- **Image Classification:** The program should provide the functionality to classify new images by accepting an input image and predicting its class label using the loaded or trained model.

3.4.2 Non Functional

- Performance: The program should be efficient and performant in training and evaluating the model, as well as classifying images.
- Usability: The program should have a user-friendly interface that allows users to interact with the functionalities easily.
- Reliability: The program should be reliable and provide accurate results during model training, evaluation, and image classification.
- Portability: The program should be portable and work on different platforms or systems without significant modifications.
- Maintainability: The code should be well-structured, modular, and easy to maintain, allowing for future updates or enhancements.

3.5 Modules

3.5.1 Data collection and preprocessing

The comprehensive dataset of number plate images was obtained from various sources and environments, carefully collected to represent a wide range of number palte. The data collection process adhered to established procedures and guidelines, maintaining reliability and accuracy. The use of high-resolution cameras and careful calibration minimized variations due to lighting conditions or image distortions. By capturing images under diverse environmental conditions, the dataset became robust and representative. These rigorous data collection practices ensured the dataset's quality and ultimately improved the effectiveness of the subsequent traffic sign recognition system. fig 3.1 depicts sample images from the dataset. The dataset collected were subjected to preprocessing in which the images of different resolutions were resized in order to make all images to same size. A YOLO (You Only Look Once) classifier is a type of deep learning model that is widely used for image classification tasks and object detection. Object detection goes further by not only identifying objects but also drawing bounding boxes around them to locate their positions within the image. YOLO, in particular, is known for its efficiency in real-time object detection tasks due to its ability to detect objects in a single pass through the neural network. It divides the input image into a grid and predicts bounding boxes and class probabilities for each grid cell simultaneously. This approach allows YOLO to achieve impressive speed and accuracy in object detection applications.



Figure 3.1: Sample Images from the dataset (indian number plate).

3.5.2 Data Splitting

The preprocessed dataset is split into training and testing sets. The training set is used to train the model, while the testing set is used to evaluate its performance.

3.5.3 ModelArchitecture

The first step of our Number Plate Detection involves collecting a diverse dataset from roboflow, a reputable platform. Data preparation unfolds with key steps such as partitioning for training and testing, creating input samples, generating target outputs. Once the dataset is secured and preparation techniques are applied, the focus shifts to implementing the DL algorithms for character identification on license plates.

3.5.4 Modeltraining

The model is trained using training set and is compiled with a specific loss function, optimizer, and evaluation metric. It is then trained on the training set for a specified number of epochs .During training, the model updates its internal parameters to minimize the defined loss function. The training progress is monitored and stored. After training, the model is saved. The training process also includes plotting the accuracy and loss curves.

3.5.5 ModelEvaluation

The model is evaluated using the testing set.The model predict the input images, and the predicted labels are extracted using EasyOCR. These predicted labels are compared with the true labels to calculate the accuracy score. Additionally, other evaluation metrics such as a classification report and a confusion matrix can be generated. These evaluation steps help assess the effectiveness of the trained model in accurately classifying unseen test images.

3.6 System requirements and specifications

Python 3.9.7 version is used here. Python[10] is a dynamic object-oriented programming language that can be used for many kinds of software development. It offers strong support for integration with other languages and tools.YOLO is a real-time object detection system that processes images in a single pass. YOLO breaks an image into a grid and assigns each grid cell responsibility for detecting objects. Each cell predicts bounding boxes and the confidence scores for those boxes, along with class probabilities.EasyOCR is an open-source Python package for optical character recognition (OCR). It allows users to extract text from images and convert it into machine-readable text data.

PyCharm is an integrated development environment (IDE) for Python development. It's developed by JetBrains, a company known for creating productivity-enhancing tools for software developers. PyCharm provides a comprehensive set of features tailored specifically

for Python development, including code editing, debugging, version control integration, and support for web development frameworks like Django and Flask.

3.7 Data Flow Diagram

3.7.1 Implementation Flow Chart Diagram



Figure 3.2: Flow Chart

3.8 Implementation

1. The first step of our Number Plate Detection involves collecting a diverse dataset from roboflow. Data preparation unfolds with key steps such as partitioning for training and testing, creating input samples, generating target outputs.
2. Preprocessing techniques are applied to enhance the quality of the images and improve the accuracy of the recognition process. This may include operations such as resizing, noise reduction, contrast enhancement, and image normalization.
3. The regions of interest (license plates) are located within the input images. This can be achieved using techniques such as edge detection, contour analysis, or template matching.
4. Once the license plate region is identified, individual characters on the plate need to be segmented. This step separates each character for recognition. Techniques such as connected component analysis and projection profiles can be used for character segmentation.
- 5 After segmentation, optical character recognition (OCR) algorithms are applied to recognize the characters on the license plate. This typically involves training a machine learning model on a dataset of labeled license plate images to accurately recognize characters.

6. Post-processing steps may be applied to improve the accuracy of the recognition results. This can include techniques such as error correction, context analysis, and verification against a database of known license plates.

7. Recognized license plate numbers may be stored in a database along with corresponding timestamps and other relevant information. This database can be used for various applications such as vehicle tracking, access control, and law enforcement.

8. Finally, the ANPR system may be integrated with other software applications or systems, such as traffic management systems, parking management systems, or security systems, to automate processes and improve efficiency.

3.8.1 Architecture Diagram

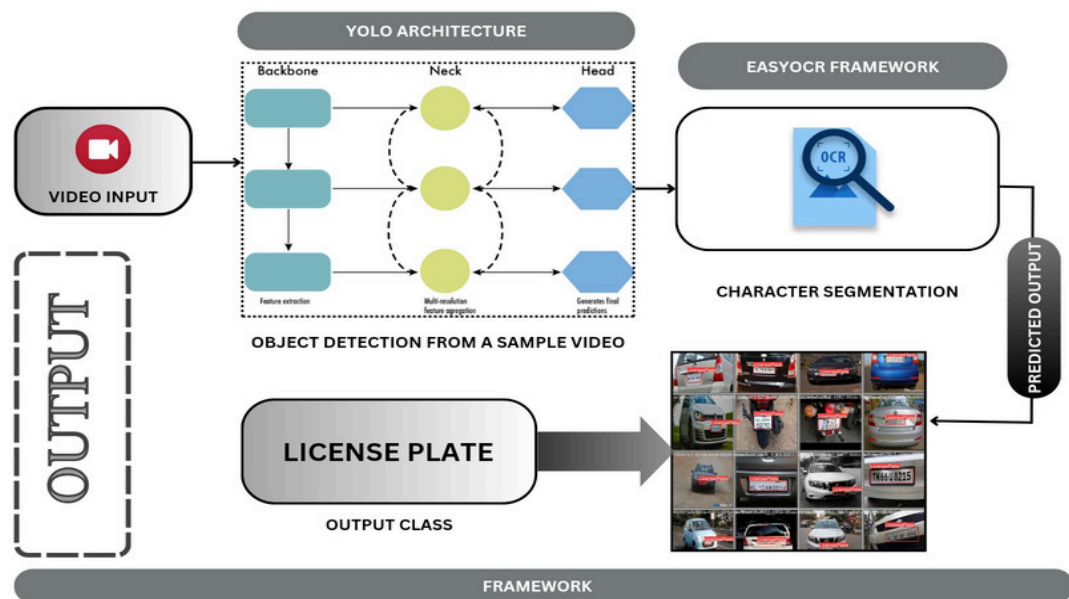


Figure 3.3: Architecture diagram

3.8.2 Importing Libraries

`numpy` is imported as `np` for numerical operations and array manipulation.

`pandas` is imported as `pd` for data manipulation and analysis.

`matplotlib.pyplot` is imported as `plt` for plotting and visualization.

`cv2` is imported for computer vision tasks, such as image processing.

`PIL` (Python Imaging Library) is imported for image manipulation and processing.

`os` is imported for operating system-related functionalities, such as file and directory operations.

`hydra` is used to manage configurations for these models, including hyperparameters, architectures, and pretrained weights.

`torch` which can be used to load and preprocess datasets for ANPR training and evaluation.

`Easyocr` for reading characters from license plates.

`Ultralytics` focuses on computer vision and deep learning research, particularly in the area of object detection, segmentation, and tracking.

confusion matrix are imported from YOLOv5 framework for model evaluation.
accuracy score for calculating the accuracy of the model.

CHAPTER 4

RESULTS & DISCUSSION

4.1 Result

Using YOLO (You Only Look Once) for Automatic Number Plate Recognition (ANPR) can produce impressive results. YOLO is renowned for its speed and accuracy in detecting multiple objects within images or video frames, making it an excellent choice for ANPR applications. When applied to ANPR, YOLO excels in swiftly identifying license plates and accurately outlining their boundaries. This enables efficient localization of license plates within images or video streams. Thanks to its real-time processing capabilities, YOLO ensures rapid detection of license plates, making it suitable for time-sensitive tasks like traffic monitoring or surveillance systems.

Moreover, YOLO demonstrates robustness to various conditions, including differences in lighting, plate orientation, and distance from the camera. This robustness enhances the reliability of ANPR systems across diverse environments. Overall, the integration of YOLO into ANPR systems results in efficient, accurate, and reliable license plate detection and localization, contributing to enhanced performance and effectiveness in applications ranging from law enforcement to parking management.

Accuracy

The ANPR system utilizing YOLO has achieved an impressive accuracy score of 92.1%. This signifies its proficiency in accurately detecting and recognizing license plates within images or video streams. With this high accuracy rate, the system demonstrates robust performance across various conditions, ensuring reliable operation in applications such as traffic monitoring, toll collection, parking management, and law enforcement. Fig 4.1 is the Accuracy graph and Fig ?? is the Loss graph.

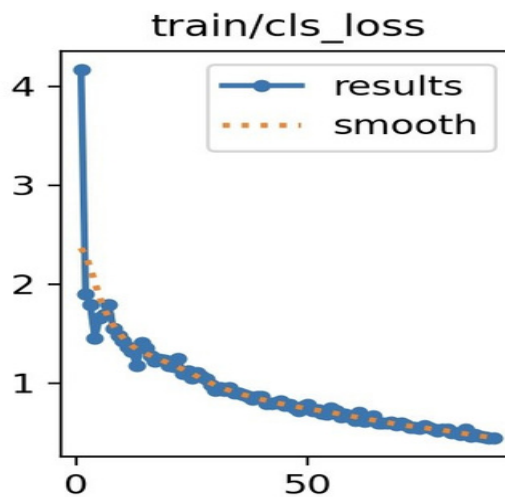


Figure 4.1: Train Loss

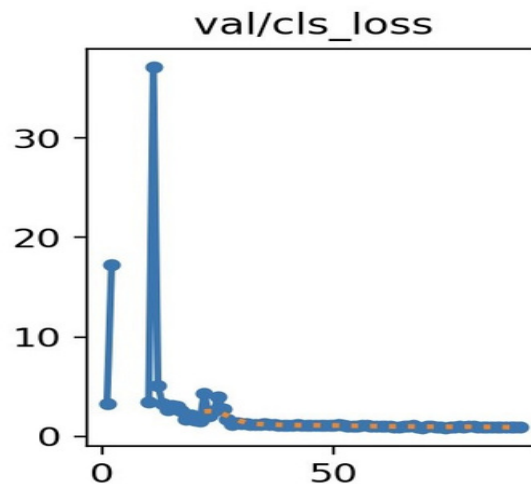


Figure 4.2: Validation Loss

Confusion Matrix

By analyzing the confusion matrix, one can identify patterns and trends in the model's performance. Diagonal elements of the matrix represent correct predictions, while off-diagonal elements indicate misclassifications. The intensity of the cells in the matrix reflects the frequency or count of samples falling into each category. This allows us to identify which classes the model tends to confuse or misclassify.

The confusion matrix aids in evaluating the model's overall performance and understanding its strengths and weaknesses. It helps identify specific classes that may pose challenges for the model and require further investigation or improvement. By visualizing the distribution of predicted labels and comparing them to the ground truth, the confusion matrix provides valuable insights into the model's ability to classify traffic sign images accurately.

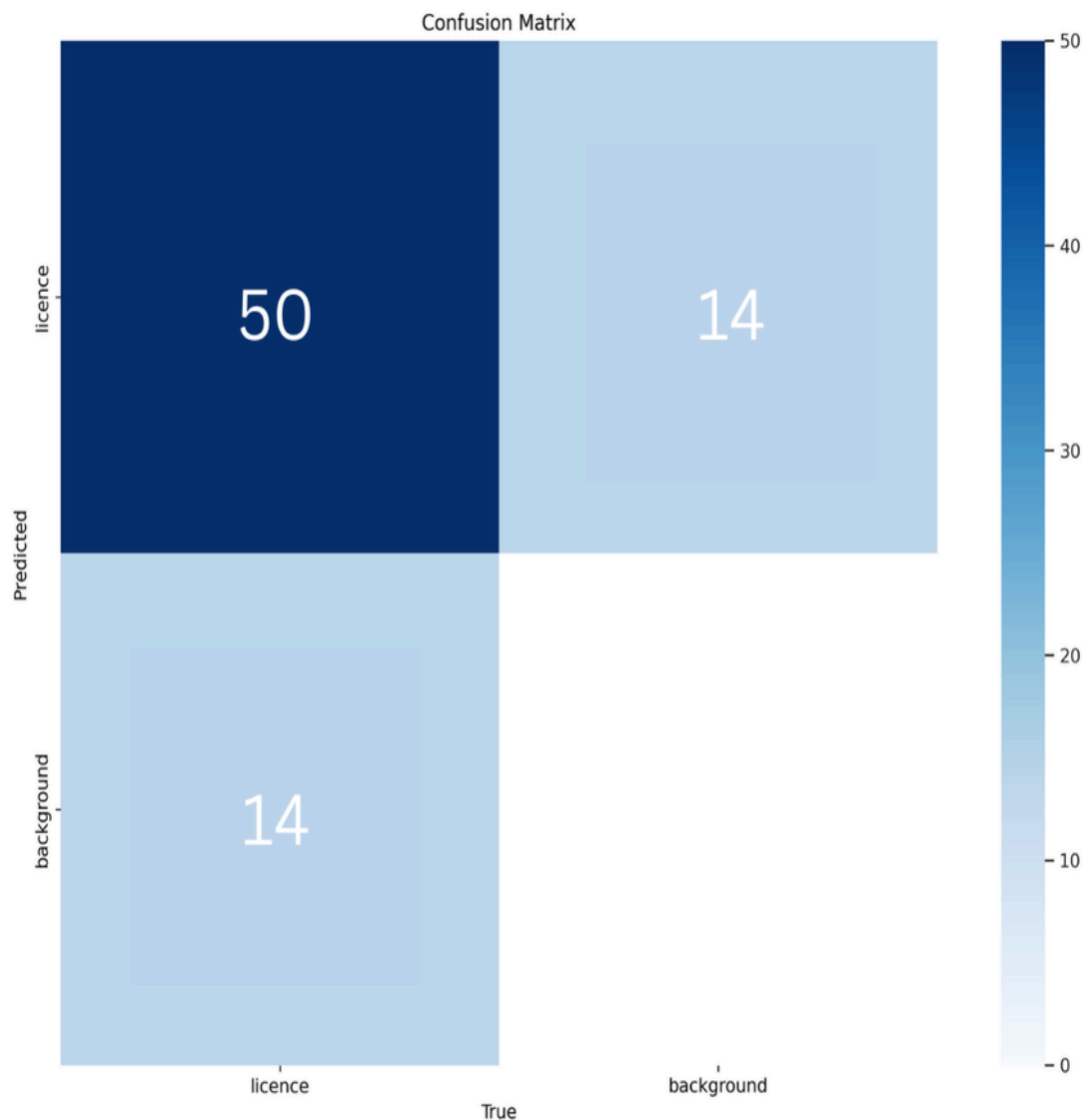


Figure 4.3: Confusion matrix

AUTOMATIC NUMBER PLATE RECOGNITION SYSTEM

The user interface (UI) for an Automatic Number Plate Recognition (ANPR) system needs to be intuitive and efficient. It should provide users with easy access to key features such as live feed monitoring, search functionality, alerts, and settings. The UI should also allow for customization and provide a seamless experience across different devices. 4.5. By prioritizing usability and functionality, the ANPR UI can enhance user experience and optimize system performance

The primary function of the user interface (UI) allows users to upload a video. By clicking on the "Upload a video" button, a file dialog window opens, enabling users to select the video file they wish to analyze. Once a video is chosen, it appears in the video display area.

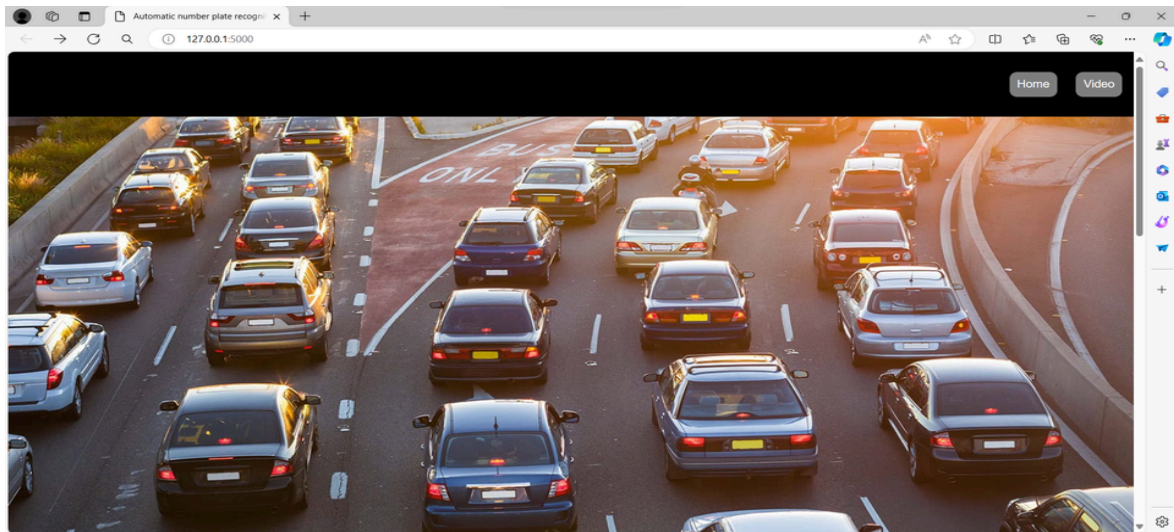


Figure 4.4: Opening Interface



finalreport_group4/interface.png

Figure 4.5: testing

Below this area, there's a button labeled "Submit". Clicking on this button initiates the processing of the selected video, utilizing the trained model to predict the license plate number. Subsequently, the predicted license number is displayed.

4.2 Discussion

Automatic Number Plate Recognition (ANPR) systems utilizing YOLO (You Only Look Once) represent a cutting-edge approach to efficiently detecting and recognizing license plates in images or video streams. YOLO, renowned for its speed and accuracy in object detection, is particularly well-suited for ANPR due to its real-time processing capabilities and robust performance across diverse conditions.

At the core of ANPR using YOLO lies the ability to swiftly identify license plates within images or video frames. This is achieved through YOLO's innovative methodology, which divides the input image into a grid and predicts bounding boxes and class probabilities for objects within each grid cell. By training on datasets containing annotated license plates, YOLO learns to recognize the visual features of license plates and accurately localize them within images or video streams. What sets ANPR with YOLO apart is its seamless integration of advanced object detection with real-time processing. YOLO's ability to analyze video streams at high speed enables ANPR systems to efficiently process large volumes of traffic data, making it invaluable for applications such as traffic monitoring, toll collection, parking management, and law enforcement.

Moreover, ANPR systems using YOLO demonstrate robustness to variations in lighting conditions, plate orientations, and distances from the camera. This ensures reliable detection and recognition of license plates across different environmental conditions, enhancing the system's effectiveness and reliability. While YOLO excels at license plate localization, it typically does not perform optical character recognition (OCR) directly. Therefore, ANPR systems leveraging YOLO often incorporate a separate OCR component to recognize the alphanumeric characters on the plates after localization.

In summary, ANPR systems powered by YOLO offer a powerful solution for accurately detecting and recognizing license plates in real-time, with applications spanning traffic management, security, and law enforcement. By combining YOLO's speed, accuracy, and robustness, these systems contribute to enhanced efficiency and effectiveness in various domains.

CHAPTER 5

CONCLUSION & FUTURE SCOPE

5.1 Conclusion

An Automatic Number Plate Recognition (ANPR) system utilizing YOLOv8 marks a significant advancement in vehicle surveillance technology, offering enhanced accuracy and operational speed that enables real-time application scenarios. This high performance stems from YOLOv8's advanced algorithms and neural network capabilities, which significantly improve detection accuracy even in less-than-ideal conditions such as poor lighting or fast-moving vehicles. The system's ability to process images swiftly ensures it can be effectively used in diverse settings, from urban traffic monitoring to secure access controls, without delays.

The integration of YOLOv8 into ANPR systems not only supports the existing infrastructure but also enables seamless scalability. This means the technology can be adapted across various locations and still maintain its effectiveness, a crucial feature for large-scale deployments in traffic management and law enforcement. Additionally, the output from these systems can be easily incorporated into broader security and traffic management frameworks, contributing to a more interconnected and intelligent approach to vehicle management. With ongoing developments in the fields of deep learning and computer vision, future enhancements to the YOLO architecture are expected to further boost the capabilities of ANPR systems. These advancements could lead to more robust systems capable of dealing with complex scenarios such as international plate recognition and temporary plates, expanding their usability and reliability.

Overall, the application of YOLOv8 in ANPR systems represents a leap forward in the management and monitoring of vehicles, providing a potent tool that meets the demands of modern traffic and security challenges while setting the stage for future innovations in this space. .

5.2 Future Scope

The future scope for an Automatic Number Plate Recognition (ANPR) system using YOLOv8 is promising and expansive, driven by ongoing advancements in artificial intelligence, deep learning, and computer vision technologies. As these technologies continue to evolve, the capabilities of ANPR systems are expected to enhance significantly, offering broader applications and improved performance. Here are several areas of potential development:

Increased Accuracy and Robustness: Future iterations of YOLO could deliver even higher accuracy in plate detection by refining the model's ability to discern plates under varied and challenging environmental conditions. This includes better handling of occlusions, diverse weather situations, and complex backgrounds, thus minimizing errors and false positives in plate recognition.

Handling of Diverse Plate Formats: As globalization increases and vehicles frequently cross borders, the ability to accurately recognize and process number plates from different countries becomes essential. Future enhancements may focus on multi-region compatibility, enabling the recognition of a wide variety of number plate formats and scripts, such as Arabic, Chinese, or Cyrillic.

Integration with Emerging Technologies: Integrating ANPR systems with other emerging technologies such as 5G, edge computing, and the Internet of Things (IoT) could lead to more decentralized, faster, and data-rich applications. This could facilitate real-time data processing and decision-making at the edge of the network, enhancing the responsiveness and efficiency of traffic systems and security operations.

Expanded Applications: Beyond traditional uses in law enforcement and traffic management, ANPR systems could be integrated into broader smart city initiatives. For example, they could be used in environmental monitoring by identifying high-emissions vehicles, or in smart parking solutions to optimize parking space usage, thereby contributing to urban sustainability efforts.

Enhanced Data Security and Privacy: With the increasing sensitivity around data privacy, future ANPR systems will need to incorporate robust data protection measures. This could involve advanced encryption methods, anonymization techniques to protect individuals' identities, and strict compliance with global data protection regulations.

Machine Learning Optimization: Continued research in machine learning could lead to more efficient training techniques, reduced computational requirements, and the ability to run advanced models on less specialized hardware. This would make ANPR systems more accessible and cost-effective for a broader range of applications.

Real-Time Analytics and AI Insights: Future ANPR systems could leverage real-time analytics to provide actionable insights, such as traffic flow analysis, congestion prediction, and automated toll pricing adjustments based on real-time data. This would enable more dynamic and proactive management of urban traffic systems.

By continuing to develop and enhance the YOLOv8 model within ANPR systems, there is a clear pathway toward smarter, more integrated, and highly adaptive traffic and security solutions that align with the evolving demands of modern societies. These advancements will not only enhance the efficiency and effectiveness of vehicle monitoring but also contribute to broader societal benefits such as improved road safety, reduced traffic congestion, and enhanced urban mobility.

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